

Max Shen

chihhans@usc.edu ◇ github.com/maxshen1212 ◇ linkedin.com/in/maxshen1212

EDUCATION

University of Southern California , Los Angeles, CA <i>M.S. in Computer Science</i>	Jan 2026 – May 2027 <i>GPA: 4.0 / 4.0</i>
National Chin-Yi University of Technology , Taichung, Taiwan <i>B.Eng. in Computer Science and Information Engineering</i>	Sept 2019 – Jun 2023

SKILLS

Programming Languages: Python, C++, TypeScript, JavaScript, Java, SQL
AI & Robotics: PyTorch, ROS2, Isaac Sim/Lab, LeRobot, TensorFlow, OpenCV
Web & Mobile: React, React Native, Express, FastAPI, RESTful APIs
DevOps & Tools: Docker, Linux, Git, GitHub Actions, CI/CD, AWS, Terraform, n8n, AI Coding Agents

RESEARCH EXPERIENCE

SLURM Lab, USC <i>Advisor: Daniel Seita</i>	Jan 2026 – Present <i>Los Angeles, CA</i>
- Formulating uncertainty estimation to benchmark robotic world models and flag inaccurate policy rollouts, validated on a physical DROID Franka platform. - Investigating diffusion- and flow-based Vision-Language-Action models for policy interpretability and generalization in robotic manipulation.	
Smart Manufacturing Lab, NCUT <i>Advisor: Yu-Chi Liu</i>	Aug 2022 – Jun 2023 <i>Taichung, Taiwan</i>
<i>Project: Adaptive Thermal Equilibrium Control System for CNC Machines</i> - Formulated a closed-loop framework coupling an MLP with MOPSO to predict and actively compensate for spindle thermal tilting ($\Delta X, \Delta Y$) at 99.1% predictive accuracy. - Suppressed thermal drift from 3–12 μm to under 1 μm on a physical 3-axis CNC vertical machining center, improving machining precision by 60%–80%.	

PROFESSIONAL EXPERIENCE

Graphen – R&D Team <i>Robotics AI Intern</i>	May 2026 – Aug 2026 <i>Taipei, Taiwan</i>
- Scaled a single-arm platform to synchronized bimanual manipulation, fine-tuning NVIDIA Isaac GR00T via Sim-to-Real imitation learning to lift policy success from 30% ID to 60% OOD. - Architected a greenfield multimodal data infrastructure integrating teleoperation pipelines and MoCap capture to scale demonstration collection. - Deployed distributed multi-GPU PyTorch training with synthetic demonstration generation via NVIDIA Cosmos to enhance out-of-distribution policy generalization.	
DoubleSpot <i>Software Engineer Intern</i>	Sep 2025 – May 2026 <i>Seattle, WA</i>
- Partnered with the CTO to redesign legacy services into a decoupled React and Express architecture, leveraging AI coding agents to accelerate feature delivery cycles by 40%. - Built GitHub Actions CI/CD pipelines and provisioned AWS infrastructure with Terraform, orchestrating ECS, ALB, RDS, and S3 for high availability in production. - Engineered a multi-tenant React Native application with tenant data isolation, onboarding and securing contracts with 20+ enterprise clients.	
Yois <i>Software Engineer</i>	Aug 2023 – Jul 2025 <i>Taipei, Taiwan</i>
- Led 5 engineers to migrate on-premise backends to AWS, syncing POS data across 10+ stores at 99.9% uptime. - Built REST ingestion pipelines cutting sync latency 40%; rebuilt the mobile app for 50% faster startup.	